FurniCraft – Requirements Gathering

An Interactive 3D Furniture E-Commerce Platform with AI Assistance · Graduation Project · DEPI · October 2026

1. Stakeholder Analysis

Stakeholder	Role	Main needs	Influence
Customer (Shopper)	Browses, customizes, and buys furniture	See the product in 3D, customize it, know the exact price, check out easily, get help finding products	High
Administrator (Store Manager)	Manages the store	Add and edit products, pricing, and inventory; view orders	High
Development Team	Designs and builds the platform	Clear requirements, realistic scope, a stable tech stack	High
Lecturer / Supervisor	Evaluates the project	Complete documentation, a working system, clear grading criteria	High
LLM API Provider	External service used by the AI assistant	Valid requests within usage limits	Low

2. User Stories

2.1 Shopping experience

US-01 MVP As a shopper, I want to browse furniture by category, room, and product type so that I can quickly find relevant items.

US-02 MVP As a shopper, I want to search, filter, and sort products so that I can narrow down the options that fit my needs.

US-03 MVP As a shopper, I want a detailed page for each product so that I can make an informed decision.

2.2 3D viewer and customization

US-04 MVP As a shopper, I want to rotate and zoom an interactive 3D model so that I can see the product from every angle.

US-05 MVP As a shopper, I want to change the color, material, and size of a product and see the 3D preview update immediately so that I can see exactly what I am buying.

US-06 MVP As a shopper, I want the total price to update live when I change an option so that I understand how my choices affect the cost.

2.3 Account, cart, and orders

US-07 MVP As a visitor, I want to create an account and log in so that my data and orders are saved.

US-08 MVP As a shopper, I want to add customized products to a cart with their selected options and adjust quantities so that I can buy exactly what I configured.

US-09 MVP As a shopper, I want to check out and place an order so that I can complete my purchase.

US-10 MVP As a shopper, I want to view my order history so that I can track what I have bought.

US-11 FUTURE As a shopper, I want a wishlist so that I can save products for later.

US-12 FUTURE As a shopper, I want to compare products side by side so that I can choose between them.

2.4 AI assistant

US-13 MVP As a shopper, I want to describe what I need in natural language (for example, "a small dark wooden desk for a bedroom under 5,000 EGP") and receive matching products so that I can find items without manual filtering.

US-14 FUTURE As a shopper, I want to describe a change in plain text (for example, "make the sofa light gray with a wooden frame") and see the 3D model update so that customizing is faster.

US-15 FUTURE As a shopper, I want clear AI-generated descriptions and style suggestions so that I can understand each product quickly.

2.5 Administration

US-16 MVP As an admin, I want to add, edit, and remove products so that the catalog stays current.

US-17 MVP As an admin, I want to manage prices, customization options, and inventory so that customers see accurate information.

US-18 FUTURE As an admin, I want AI-generated product descriptions based on product data so that the catalog stays consistent and quick to expand.

US-19 MVP As an admin, I want to manage orders so that I can follow each purchase.

US-20 MVP As an admin, I want simple reports and analytics so that I can understand the store activity.

3. Use Cases

3.1 Use case summary

ID	Use case	Primary actor	Priority
UC-01	Browse and search products	Shopper	MVP
UC-02	View product details	Shopper	MVP
UC-03	Customize a product in 3D	Shopper	MVP
UC-04	Customize a product using natural language	Shopper, AI Assistant	Future
UC-05	Ask the AI assistant to find products	Shopper, AI Assistant	MVP
UC-06	Register and log in	Visitor	MVP
UC-07	Manage cart	Shopper	MVP
UC-08	Checkout and place order	Shopper	MVP
UC-09	View order history	Shopper	MVP
UC-10	Manage wishlist and compare products	Shopper	Future
UC-11	Manage products, pricing, inventory, and orders	Admin	MVP
UC-12	Generate product descriptions	Admin, AI Assistant	Future

3.2 Detailed use cases

UC-03: Customize a product in 3D

- **Actor:** Shopper
- **Precondition:** The product page is open and the 3D model has loaded.

- **Main flow:** (1) The shopper rotates and zooms the model. (2) The shopper selects a color, material, or size. (3) The system updates the 3D preview. (4) The system recalculates and displays the total price. (5) The shopper adds the customized product to the cart.
- **Alternative flow:** If the selected combination is not available, the system shows a message and keeps the previous valid selection.
- **Postcondition:** The selected customization is stored with the cart item.

UC-04: Customize a product using natural language (Future)

- **Actors:** Shopper, AI Assistant
- **Precondition:** A product is open in the 3D viewer.
- **Main flow:** (1) The shopper types a request, for example "make the sofa light gray with a wooden frame". (2) The system sends the request to the AI assistant. (3) The AI returns structured JSON (color, material, size). (4) The system validates the JSON against the available options. (5) The 3D viewer applies the changes and the price updates.
- **Alternative flow:** If the request cannot be understood or the options are unavailable, the assistant explains what is available and asks the shopper to rephrase.
- **Postcondition:** The viewer shows the requested customization.

UC-05: Ask the AI assistant to find products

- **Actors:** Shopper, AI Assistant
- **Main flow:** (1) The shopper writes a request with constraints such as type, room, color, material, and budget. (2) The assistant interprets the request. (3) The system returns matching products from the catalog.
- **Alternative flow:** If nothing matches, the assistant suggests the closest available products.
- **Postcondition:** The shopper sees a list of relevant products. Answers are limited to available products, options, and prices.

UC-08: Checkout and place order

- **Actor:** Shopper
- **Precondition:** The shopper is logged in and has at least one item in the cart.
- **Main flow:** (1) The shopper reviews the cart. (2) The shopper enters the required order details. (3) The shopper pays through a payment gateway in test mode or a mock payment. (4) The system confirms the order. (5) The order appears in the order history.
- **Alternative flow:** If the shopper is not logged in, the system redirects to login and then returns to checkout.
- **Postcondition:** The order is saved and inventory is updated.

4. Functional Requirements

4.1 Shopping experience

FR-01 MVP The system shall let users browse products by category, room, and product type.

FR-02 MVP The system shall provide advanced search with precise filters, and sorting of products.

FR-03 MVP The system shall provide a detailed page for each product.

4.2 3D viewer and customization

FR-04 MVP The system shall display interactive 3D models with full rotation and zoom.

FR-05 MVP The system shall let users select color, material, and size in real time.

FR-06 MVP The system shall update the 3D preview immediately after each selection.

FR-07 MVP The system shall recalculate and display the total price live after each selection.

FR-22 MVP Each 3D model shall have separate parts (for example, the frame and the fabric) so that color and material can be changed for each part.

4.3 E-commerce functionality

FR-08 MVP The system shall provide a cart that stores each item with its saved customization and supports quantity management.

FR-09 MVP The system shall provide checkout, with the payment gateway in test mode or a mock payment.

FR-10 MVP The system shall provide user registration, login, and user accounts.

FR-11 MVP The system shall provide order history for each user.

FR-12 FUTURE The system shall provide a wishlist.

FR-13 FUTURE The system shall provide product comparison (for example by price, model, and color).

4.4 Admin panel (simplified)

FR-14 MVP The system shall let admins create, edit, and delete products.

FR-15 MVP The system shall let admins manage pricing and customization options.

FR-16 MVP The system shall let admins manage inventory.

FR-23 MVP The system shall let admins manage orders.

FR-24 MVP The system shall provide simple reports and analytics for admins.

4.5 AI component

FR-17 MVP The system shall provide an AI shopping assistant that understands natural-language requests and recommends matching products.

FR-18 FUTURE The system shall convert natural-language customization requests into structured JSON (color, material, size) that the 3D viewer applies automatically.

FR-19 FUTURE The system shall generate product descriptions and style suggestions from product data.

FR-20 MVP The system shall limit AI answers to available products, options, and prices (guardrails).

FR-21 MVP The system shall validate AI output before applying it to the viewer or the database.

5. Non-Functional Requirements

The numeric targets below are proposed starting points and will be validated during testing.

ID	Category	Requirement
NFR-01	Performance	Pages should load quickly; the proposed target is under 3 seconds on a normal connection.
NFR-02	Performance	The 3D viewer should stay smooth during rotation and zoom, and price updates should appear without noticeable delay after a selection.
NFR-03	Performance	AI responses should return in a reasonable time, and the UI should show a loading state while waiting.
NFR-04	Security	User authentication shall use JWT, and protected routes and admin functions shall require authorization.

ID	Category	Requirement
NFR-05	Security	User input and AI output shall be validated; passwords shall not be stored in plain text.
NFR-06	Usability	The interface shall be clear and consistent, and the customization flow shall be easy to follow without instructions.
NFR-07	Usability	The platform shall be responsive and work on desktop and mobile screens.
NFR-08	Reliability	The system shall handle errors gracefully, including failed AI calls and unavailable 3D assets, with clear messages to the user.
NFR-09	Scalability	The architecture shall allow new products, models, and options to be added without code changes to the core system.
NFR-10	Maintainability	The code shall be modular and documented, with a RESTful API separating frontend and backend.